

Unit - III	PARTICULAR	Marks
	Understanding Controls Net Common Controls: Control Hierarchy, Label and Link Label, Text Box and Rich Text Box, Picture Box, Button, Group Box, Panel, Check Box and Radio Button, List Box, Checked List Box and Combo Box, Month Calendar and Date Time Picker, Tree View and List View, Timer, Track Bar and Progress Bar, Image List control, HScroll Bar, VScroll Bar, Tab Control. Common Dialogs Control: Color Dialog, Folder Browser Dialog, Font Dialog, Open File Dialog, Save File Dialog , MDI-Forms, Exploring Properties, Methods and Events, Menu bar, Context Menu, Message box, Input box.	

Unit – III : Understanding .NET Common Controls

Control Hierarchy

- All controls in ASP.NET have a hierarchy (parent-child relationship).
- Example: Page →Panel →Button.
- Helps organize controls systematically.

Property Name	Description
ID	Unique name for the control.
Controls	Collection of child controls.
Parent	Returns parent container of control.

Label Control

- Displays read-only text on a form.
- Used for headings, captions, or instructions.
- User cannot edit the label text.

Property Name	Description
Text	Text shown in label.
ForeColor	Sets label text color.
Font	Sets font style and size.

LinkLabel (HyperLink)

- Displays clickable link text.
- Useful for navigation to websites, files, or forms.

- Looks like a web hyperlink.

Property Name	Description
Text	Text of link.
NavigateUrl	Target URL to open.
Target	Defines how to open (_blank, _self).

TextBox Control

- Used for user text input.
- Can be single-line or multi-line.
- Example: Username, Password input.

Property Name	Description
Text	Gets/sets user input text.
TextMode	SingleLine, MultiLine, Password.
MaxLength	Maximum characters allowed.

RichTextBox Control (MultiLine TextBox)

- Similar to TextBox but allows multi-line content.
- Used for writing paragraphs or notes.
- Example: Comment Box.

Property Name	Description
Rows	Number of visible rows.
Columns	Number of visible columns.
Wrap	Word wrapping enabled/disabled.

PictureBox (Image)

- Displays an image (JPG, PNG, GIF).
- Used for logos, profile pictures, etc.
- Image is static (not editable).

Property Name	Description
ImageUrl	Path of image file.
AlternateText	Text shown if image not available.
Height / Width	Size of image.

Button Control

- Executes an action when clicked.
- Example: Submit, Cancel, Login.
- Causes a postback to server.

Property Name	Description
Text	Caption of button.

Enabled	Enables/disables button.
BackColor	Sets background color.

GroupBox (Panel with GroupingText)

- Groups related controls inside a box.
- Provides a border with title text.
- Example: Gender Selection group.

Property Name	Description
GroupingText	Title text of group.
BorderStyle	Style of border.
BackColor	Background color.

Panel Control

- Container control for grouping.
- Can have scrollbars if content is large.
- Helps layout design.

Property Name	Description
ScrollBars	Adds scrollbars.
BorderStyle	Sets border style.
BackColor	Background color.

CheckBox Control

- Represents Yes/No selection.
- User can select multiple checkboxes.
- Useful for conditions or multi-options.

Property Name	Description
Text	Caption text.
Checked	True/False if selected.
AutoPostBack	Triggers immediate postback.

RadioButton Control

- Allows user to select only one option.
- Example: Gender (Male/Female).
- GroupName links options together.

Property Name	Description
Text	Caption text.

GroupName	Groups radio buttons.
Checked	True/False if selected.

ListBox Control

AppleBanana

- Displays a list of items.
- User can select one or multiple.
- Example: Select subjects.

Property Name	Description
Items	Collection of list items.
SelectionMode	Single or Multiple selection.
SelectedValue	Gets selected item value.

CheckedListBox (CheckBoxList)

- List of items with checkboxes.
- User can select multiple items.
- Example: Select languages known.

Property Name	Description
Items	List of items.
SelectedIndex	Index of selected item.
RepeatDirection	Horizontal/Vertical arrangement.

ComboBox (DropDownList)

RedBlue

- Drop-down list of options.
- Saves space on form.
- Only one item selectable.

Property Name	Description
Items	Collection of list items.
SelectedValue	Value of chosen item.
AutoPostBack	Causes postback when changed.

MonthCalendar (Calendar)

- Displays a calendar view.
- User can pick a date.
- Example: Select DOB.

Property Name	Description
SelectedDate	Gets/sets chosen date.
VisibleDate	Month/year currently displayed.

SelectionMode	Day/Week/Month selection.
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DateTimePicker (CalendarExtender)

- Allows user to select date/time.
- More compact than calendar.
- Example: Pick appointment time.

Property Name	Description
SelectedDate	Gets selected date.
Format	Date format (dd/MM/yyyy).
MinDate/MaxDate	Restricts range of dates.

TreeView Control

- Displays data in hierarchical nodes.
- Example: File explorer style.
- Supports expand/collapse.

Property Name	Description
Nodes	Collection of tree nodes.
SelectedNode	Currently selected node.
ShowCheckBoxes	Adds checkboxes to nodes.

ListView Control

- Displays list of data with templates.
- Supports custom layout.
- Example: Product catalog.

Property Name	Description
Items	Collection of list items.
DataSource	Binds external data.
SelectedIndex	Index of selected item.

Timer Control

- Executes actions at regular intervals.
- Example: Refresh data every 2 sec.
- Works with UpdatePanel for Ajax.

Property Name	Description
Interval	Time in milliseconds.
Enabled	Starts/stops timer.
OnTick	Event executed on interval.

TrackBar (SliderExtender)

- Allows user to select a value by sliding.
- Example: Volume control.
- Graphical slider interface.

Property Name	Description
Minimum	Lowest value.
Maximum	Highest value.
Value	Current slider position.

ProgressBar (UpdateProgress)

Loading...
Shows progress of a process.
Example: File upload progress.
Helps user know waiting time.

Property Name	Description
AssociatedUpdatePanelID	Binds with UpdatePanel.
ProgressTemplate	Content shown while loading.
DisplayAfter	Delay before showing progress.

ImageList (Simulated)

- Stores collection of images.
- Used in menus, buttons, or galleries.
- Not direct in ASP.NET, but simulated.

Property Name	Description
ImageUrl	Path of image.
Height/Width	Size of image.
AlternateText	Alternate text.

HScrollBar (Panel with Scroll)

- Provides horizontal scrolling.
- Used for wide content.
- Example: Viewing a large image.

Property Name	Description
ScrollBars	Horizontal/Vertical/Auto.
Width	Control width.
Height	Control height.

VScrollBar (Panel with Scroll)

- Provides vertical scrolling.
- Used for long content.
- Example: Chat messages panel.

Property Name	Description
ScrollBars	Vertical/Horizontal/Auto.
Height	Control height.
Width	Control width.

TabControl (TabContainer)

First Tab ContentSecond Tab Content

- Displays multiple pages in tab style.
- Example: Browser-like tabs.
- Saves space with organized content.

Property Name	Description
HeaderText	Title of each tab.
ActiveTabIndex	Shows currently selected tab.
TabPanels	Collection of all tabs.

CommonDialogs&Other Controls

Color Dialog

Used to choose a color from color palette.

Example: Change text/background color.

Can return selected color value.

Property Name	Description
Color	Gets/sets selected color.
AllowFullOpen	Allows user to open custom colors.
AnyColor	Show all available colors.

Folder Browser Dialog

```
FolderBrowserDialog fbd = new FolderBrowserDialog(); fbd.ShowDialog();
```

Used to select a folder path.

Common in saving/exporting files.

Returns selected folder path as string.

Property Name	Description
SelectedPath	Returns chosen folder path.
RootFolder	Starting location of dialog.
Description	Text shown in dialog.

Font Dialog

```
FontDialog fd = new FontDialog(); fd.ShowDialog();
```

Used to pick font style, size, and color.

Common in editors (Word, Notepad).

Returns font object.

Property Name	Description
Font	Gets/sets selected font.
ShowColor	Allows font color selection.
Color	Gets/sets selected color.

Open File Dialog

```
OpenFileDialog ofd = new OpenFileDialog(); ofd.ShowDialog();
```

Lets user select a file to open.

Example: Open Word, Image, Text file.

Can apply filters.

Property Name	Description
FileName	Selected file path.
Filter	File type filter (e.g., *.txt).
Multiselect	Allows selecting multiple files.

Save File Dialog

```
SaveFileDialog sfd = new SaveFileDialog(); sfd.ShowDialog();
```

Lets user save a file.

Common in editors, image apps.

Returns save path.

Property Name	Description
FileName	Path to save file.
Filter	File extension filter.
DefaultExt	Default file extension.

MDI Forms (Multiple Document Interface)

```
Form child = new Form(); child.MdiParent = this; child.Show();
```

Allows multiple forms inside one parent form.

Example: MS Word multiple docs.

Improves multitasking.

Property Name	Description
IsMdiContainer	True if form is parent.
MdiChildren	Returns child forms.
ActiveMdiChild	Currently active child form.

Exploring Properties, Methods, Events

```
Example(Button)-Property:Text='Submit'|Method:Button1.Focus()| Event: Button1_Click()
```

Property = Controls characteristics (Text, Size).

Method = Performs action (Show(), Hide()).

Event = Triggered on action (Click, Load).

Type	Description
Property	Defines characteristics (Text, Size, Font).
Method	Action performed by control (Show, Hide).
Event	Response to user action (Click, Change).

Menu Bar

Creates a top-level menu.
Example: File, Edit, Help.
Organizes commands.

Property Name	Description
Items	Menu items collection.
Orientation	Horizontal/Vertical.
StaticDisplayLevels	Levels visible always.

Context Menu (Right-click Menu)

ContextMenuStrip cms = new ContextMenuStrip(); cms.Items.Add("Copy"); cms.Items.Add("Paste");
Appears when user right-clicks.
Example: Copy, Paste menu.
Context-based options.

Property Name	Description
Items	Menu items.
SourceControl	Control linked with context menu.
ShowCheckMargin	Shows checkboxes.

MessageBox

MessageBox.Show("Hello Students!");
Displays popup message.
Can have buttons (OK, Yes/No).
Example: Warnings, Info, Confirmations.

Property Name	Description
Text	Message content.
Caption	Title of message box.
Buttons	Type of buttons (OK, Yes/No).

InputBox

Dim name As String = InputBox("Enter your name:")
Used to take input from user in a popup.
Example: Enter name, email.
Available in VB.NET.

Property Name	Description
Prompt	Message shown to user.
DefaultResponse	Default input text.
Title	Title of input box.

